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Toothed portions engagement adaptation mechanism and logistics system

Abstract

The present disclosure relates to a toothed portions engagement adaptation mechanism and a logistics system. The toothed portions engagement adaptation mechanism includes: a rack including a first side and a second side arranged opposite to each other; a first guide tooth provided on the first side; a second guide tooth movably provided on the first side, the second guide tooth being farther than the first guide tooth from the second side; and a gear configured to partially engage the rack from the first side, and move towards the second side so as to completely engage the rack, wherein at least one of the first guide tooth and the second guide tooth is configured to engage the gear during the movement of the gear from the first side toward the second side.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This disclosure is a U.S. National Stage Application under 35 U.S.C. § 371 of International Patent Application No. PCT/CN2021/087624, filed on Apr. 16, 2021, which is based on and claims priority to CN Application No. 202010346293.4, filed on Apr. 27, 2020, the disclosures of which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

(2) The present disclosure relates to the field of mechanical transmission, and in particular to a toothed portions engagement adaptation mechanism and a logistics system.

BACKGROUND

(3) In a logistics system, a logistics vehicle needs to frequently switch from horizontal motion to vertical climbing motion. The logistics vehicle generally climbs along a rack by engagement between a gear and the rack. In the process of guiding the gear into the rack, there is a phenomenon that the gear abuts against a side of the rack and cannot move axially, which leads to a failure of axial guiding of the gear into the rack, causing a system failure and affecting the operation efficiency of the whole system.

SUMMARY

(4) In some embodiments of the present disclosure, a toothed portions engagement adaptation mechanism is provided, which includes: a rack including a first side and a second side arranged opposite to each other; a first guide tooth provided on the first side; a second guide tooth movably provided on the first side, the second guide tooth being farther than the first guide tooth from the second side; and a gear configured to partially engage the rack from the first side, and move towards the second side so as to completely engage the rack, wherein at least one of the first guide tooth and the second guide tooth is configured to engage the gear during the movement of the gear from the first side toward the second side.

(5) In some embodiments, the second guide tooth is configured to move towards the second side

under an action of an external force.

(6) In some embodiments, the toothed portions engagement adaptation mechanism further includes a mounting member, the mounting member being movably provided on the first side, and the second guide tooth being provided on the mounting member.

(7) In some embodiments, the toothed portions engagement adaptation mechanism further includes an elastic element provided between the mounting member and the first side.

(8) In some embodiments, the toothed portions engagement adaptation mechanism further includes a first shaft, the first shaft connecting the mounting member and the rack, the first shaft extending parallel to a connecting line between the first side and the second side, the mounting member being configured to translate relative to the rack.

(9) In some embodiments, the first side is provided with a first recess, the first recess being recessed towards the second side, the mounting member being arranged in the first recess, and the mounting member including an outer side flush with the first side.

(10) In some embodiments, the first guide tooth is provided in the first recess, and the mounting member is provided with an avoidance portion for avoiding the first guide tooth.

(11) In some embodiments, the toothed portions engagement adaptation mechanism further includes a second shaft, the second shaft connecting the mounting member and the rack, the second shaft extending in a direction perpendicular to a connecting line between the first side and the second side, the mounting member being configured to swing relative to the rack.

(12) In some embodiments, the first side is provided with a second recess, and the mounting member includes a first mounting portion and a second mounting portion, the first mounting portion being arranged in the second recess and the second guide tooth being arranged in the first mounting portion, the second shaft connecting the second mounting portion and the rack.

(13) In some embodiments, a back face of the rack is provided with a third recess, the third recess being communicated with the second recess, the second mounting portion being arranged in the third recess, the mounting member including outer sides flush with the back face and the first side of the rack.

(14) In some embodiments, the first guide tooth and the second guide tooth are respectively located on two sides of a central axis of the gear.

(15) In some embodiments, there is a first distance between the farthest part of the first guide tooth from the second side and the farthest part of the second guide tooth from the second side, and the second guide tooth is configured to move towards the second side under an action of an external force, a maximum distance of the movement being a second distance, the second distance being greater than the first distance.

(16) In some embodiments, a side of the first guide tooth away from the second side is configured as a first pointed shape portion, and a side of the second guide tooth away from the second side is configured as a second pointed shape portion, the first pointed shape portion and the second pointed shape portion both extending far away from the second side, and a part of the gear for initial engagement with the rack being configured as a third pointed shape portion, the first pointed shape portion and the second pointed shape portion being both adapted to the third pointed shape portion.

(17) In some embodiments, the first pointed shape portion includes a first bevel face, a second bevel face and a first end face, the first end face formed at a junction between the first bevel face and the second bevel face; and the second pointed shape portion includes a third bevel face, a fourth bevel face and a second end face, the second end face formed at a junction between the third bevel face and the fourth bevel face.

(18) In some embodiments, the rack is provided with two or more first teeth along a lengthwise extending direction, the first guide teeth being provided on a side end face of one of the two or more first teeth.

(19) In some embodiments, the rack is provided with two or more first teeth along a lengthwise

extending direction, a top of the first guide tooth and a top of the second guide tooth being both flush with tops of the two or more first teeth.

(20) In some embodiments of the present disclosure, a logistics system is provided, which includes a logistics vehicle, a frame and the toothed portions engagement adaptation mechanism described above, wherein the gear of the toothed portions engagement adaptation mechanism is provided on the logistics vehicle, and the rack of the toothed portions engagement adaptation mechanism is provided on the frame, and the logistics vehicle is configured to move along the rack by engagement of the gear and the rack.

(21) In some embodiments, the logistics system further includes a third shaft and a power mechanism provided on the logistics vehicle, wherein the third shaft connects the power mechanism and the gear, and the power mechanism is configured to drive the gear to move from the first side towards the second side of the rack.

(22) Based on the above technical solutions, the present disclosure has at least the following beneficial effects:

(23) In some embodiments, both the first guide tooth and the second guide tooth are provided, and when the gear initially engages the rack, the second guide tooth is closer than the first guide tooth to the gear, and the second guide tooth can move relative to the rack, and the first guide tooth and the second guide tooth cooperate with each other to alleviate various disadvantage factors for guiding the gear into the rack and facilitate successful guiding of the gear into the rack at all angles to engage the rack.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Drawings illustrated herein are used for providing further understanding of the present disclosure and form part of the present application, and illustrative embodiments of the present disclosure and description thereof are intended for explaining instead of improperly limiting the present disclosure. In the drawings:

(2) FIG. 1 is a schematic diagram of a toothed portions engagement adaptation mechanism provided according to some embodiments of the present disclosure;

(3) FIG. 2 is a schematic diagram of a rack and a first guide tooth provided according to some embodiments of the present disclosure;

(4) FIG. 3 is a schematic diagram of a mounting member and a second guide tooth provided according to some embodiments of the present disclosure;

(5) FIG. 4 is a schematic diagram of a gear provided according to some embodiments of the present disclosure;

(6) FIG. 5 is a schematic diagram of initial engagement between a gear and a first side of a rack provided according to some embodiments of the present disclosure;

(7) FIG. 6 is a schematic diagram of a toothed portions engagement adaptation mechanism provided according to some other embodiments of the present disclosure; and

(8) FIG. 7 is a schematic diagram of a rack and a mounting member in FIG. 6.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(9) Technical solutions in the embodiments will be described below clearly and completely in conjunction with the accompanying drawings in the embodiments of the present disclosure. Obviously, the described embodiments are only part of, instead of all of embodiments of the present disclosure. Based on the embodiments of the present disclosure, all of other embodiments obtained by a person of ordinary skill in the art without creative work should fall into the protection scope of the present disclosure.

(10) In description of the present disclosure, it needs to be appreciated that orientation or position

relations denoted by the terms “center”, “longitudinal”, “transverse”, “front”, “rear”, “left”, “right”, “vertical”, “horizontal”, “top”, “bottom”, “inner”, “outer” and the like are orientation or position relations illustrated based on the drawings, are merely for the convenience of describing the present disclosure and simplifying description, instead of indicating or implying the denoted devices or elements must have specific orientations or be constructed and operated in specific orientations, and thus the terms cannot be construed as limiting the protection scope of the present disclosure.

(11) A gear is provided on a logistics vehicle, and a rack is provided on a frame, and the logistics vehicle moves along the rack by engagement of the gear and the rack. During guiding of the gear on the logistics vehicle into the rack, there is a great possibility that a tooth end face on the gear is staggered with a tooth end face on the rack, and when the gear is axially pushed into the rack, a tooth on the gear engages a tooth on the rack and rotates by a certain angle, thus achieving successful engagement of the gear and the rack. However, there is also a situation where the tooth end face on the gear is just aligned with the tooth end face on the rack, and the gear abuts against a side of the rack and cannot move axially, resulting in a failure of axial guiding of the gear into the rack.

(12) Based on this, in some embodiments of the present disclosure, a toothed portions engagement adaptation mechanism is provided to alleviate the problem of the failure of axial guiding of the gear into the rack.

(13) As shown in FIG. 1, in some embodiments, the toothed portions engagement adaptation mechanism includes a rack **1**, a first guide tooth **2**, a second guide tooth **3**, and a gear **4**.

(14) The rack **1** includes a first side **11** and a second side **12** arranged opposite to each other. A front face of the rack **1** is provided with two or more first teeth **13** along a lengthwise extending direction of the rack **1**, and a face opposite to the front face of the rack **1** is a back face **14** of the rack **1**. A direction between the first side **11** and the second side **12** is an extending direction of each first teeth **13** of the two or more first teeth **13**.

(15) The first guide tooth **2** is provided on the first side **11** of the rack **1**.

(16) The second guide tooth **3** is movably provided on the first side **11** of the rack **1**, and the second guide tooth **3** is farther than the first guide tooth **2** from the second side **12**.

(17) The gear **4** is circumferentially provided with two or more second teeth **42**. The second teeth **42** are configured to engage the first teeth **13** on the rack **1**. The gear **4** is configured to initially engage the rack **1** from the first side **11**, move towards the second side **12** and further engage the rack **1**, so as to completely engage the rack **1**.

(18) At least one of the first guide tooth **2** and the second guide tooth **3** is configured to engage the gear **4** during the movement of the gear **4** from the first side **11** toward the second side **12**.

(19) In some embodiments, as shown in FIG. 5, during guiding of the gear **4** into the rack **1**, there are the following situations when the gear **4** initially engages the rack **1** from the first side **1** of the rack **1**.

(20) If an end face of one of the second teeth **42** on the gear **4** is located between an end face of the second guide tooth **3** and an end face of the first guide tooth **2**, as the second guide tooth **3** is farther than the first guide tooth **2** from the second side **12**, the second guide tooth **3** comes into contact with the gear **4** first, and the gear **4** rotates, under an action of the second guide tooth **3**, to a position of correct engagement with the rack **1**, thus correctly guiding the gear **4** into the rack **1**.

(21) If the end face of one of the second teeth **42** on the gear **4** is aligned with the end face of the second guide tooth **3**, such that the gear **4** abuts against the second guide tooth **3** and cannot rotate, as an axial force on the gear **4** increases, the second guide tooth **3** moves, under force, relative to the first side **11**, and when the end face of the second guide tooth **3** is farther than the end face of the first guide tooth **2** from the gear **4**, the end face of the first guide tooth **2** starts to approach an end face of another second tooth **42** on the gear **4**, and turns the gear **4** by a certain angle after contact, such that the gear **4** rotates to a position of correct engagement with the rack **1**, thus correctly guiding the gear **4** into the rack **1**.

(22) If the end face of one of the second teeth **42** on the gear **4** is aligned with the end face of the first guide tooth **2**, since the second guide tooth **3** comes into contact with the gear **4** earlier than the first guide tooth **2**, and the end face of the second guide tooth **3** is not aligned with an end face of another corresponding second tooth **42** on the gear **4** due to a geometric relationship, the second guide tooth **3** can still turn the gear **4** by a certain angle, such that the gear **4** rotates to a position of correct engagement with the rack **1**, thus correctly guiding the gear **4** into the rack **1**.

(23) Therefore, in some embodiments provided in the present disclosure, both the first guide tooth **2** and the second guide tooth **3** are provided, and the first guide tooth **2** and the second guide tooth **3** cooperate with each other to alleviate various disadvantage factors for guiding the gear **4** into the rack **1** and facilitate successful guiding of the gear **4** into the rack **1** at all angles to engage the rack **1**.

(24) In some embodiments, the first guide tooth **2** and the second guide tooth **3** are respectively located two sides of a central axis of the gear **4**.

(25) Optionally, the first guide tooth **2** is located above the central axis of the gear **4**, and the second guide tooth **3** is located below the central axis of the gear **4**. Alternatively, the first guide tooth **2** is located below the central axis of the gear **4**, and the second guide tooth **3** is located above the central axis of the gear **4**.

(26) In some embodiments, the second guide tooth **3** is adjacent to the first guide tooth **2**.

(27) In some embodiments, the second guide tooth **3** is configured to move towards the second side **12** under an action of an external force.

(28) In some embodiments, there is a first distance *a* between the farthest part of the first guide tooth **2** from the second side **12** and the farthest part of the second guide tooth **3** from the second side **12**, and the second guide tooth **3** is configured to move towards the second side **12** under the action of an external force, a maximum distance of the movement being a second distance *b*, the second distance *b* being greater than the first distance *a*.

(29) When the end face of one of the second teeth **42** on the gear **4** is aligned with the end face of the second guide tooth **3**, such that the gear **4** abuts against the second guide tooth **3** and cannot rotate, as the axial force on the gear **4** increases, the second guide tooth **3** moves, under force, in the direction toward the second side **12**, and since the second distance *b* is greater than the first distance *a*, thus when the second guide tooth **3** moves towards the second side **12** such that the end face of the second guide tooth **3** is farther than the end face of the first guide tooth **2** from the gear **4**, the end face of the first guide tooth **2** starts to approach the end face of another second tooth **42** on the gear **4** and turns the gear **4** by a certain angle after contact, such that the gear **4** rotates to a position of correct engagement with the rack **1**, thus achieving correct guiding of the gear **4** into the rack **1**.

(30) In some embodiments, as shown in FIG. 2, a side of the first guide tooth **2** away from the second side **12** is configured as a first pointed shape portion **21**. As shown in FIG. 3, a side of the second guide tooth **3** away from the second side **12** is configured as a second pointed shape portion **31**. The first pointed shape portion **21** and the second pointed shape portion **31** both extend away from the second side **12**. As shown in FIG. 4, a part of the gear **4** for initial engagement with the rack **1** is configured as a third pointed shape portion **41**, the first pointed shape portion **21** and the second pointed shape portion **31** being both adapted to the third pointed shape portion **41**.

(31) As shown in FIG. 4, the tooth end face of each second tooth **42** of the two or more second teeth **42** circumferentially provided on the gear **4** is provided with a third pointed shape portion **41**.

(32) During guiding of the gear **4** into engagement with the rack **1**, at least one of the first pointed shape portion **21** and the second pointed shape portion **31** is engaged with and guided to the third pointed shape portion **41**, and the gear **4** can passively rotate without power. In the process where the gear **4** axially approaches the rack **1** and is guided into the rack **1**, the gear **4** is axially pushed into the rack **1**, and the gear **4** passively rotates by a certain angle under the action of bevel faces of the first pointed shape portion **21** and the second pointed shape portion **31**, thus achieving correct

engagement of the gear **4** and the rack **1**.

(33) When the third pointed shape portion **41** of the gear **4** is just aligned with the first pointed shape portion **21** or the second pointed shape portion **31**, and the gear **4** abuts against the side of the rack **1** and cannot move axially, the first guide tooth **2** and the second guide tooth **3** cooperate to overcome disadvantage factors for guiding the gear **4** into the rack **1** and facilitate successful guiding of the gear **4** into the rack **1** at all angles.

(34) In some embodiments, as shown in FIG. 2, the first pointed shape portion **21** includes a first bevel face **211**, a second bevel face **212** and a first end face **213**, the first bevel face **211** and the second bevel face **212** being connected to the first end face **213** at an end close to each other.

(35) The first bevel face **211** and the second bevel face **212** respectively extend from ends of two tooth faces of a first tooth **13**, and extending ends of the first bevel face **211** and the second bevel face **212** are close to each other and connected to the first end face **213**.

(36) In some embodiments, as shown in FIG. 3, the second pointed shape portion **31** includes a third bevel face **311**, a fourth bevel face **312** and a second end face **313**, the third bevel face **311** and the fourth bevel face **312** being connected to the second end face **313** at an end close to each other.

(37) In some embodiments, as shown in FIG. 4, the third pointed shape portion **41** includes a fifth bevel face **411**, a sixth bevel face **412** and a third end face **413**, the fifth bevel face **411** and the sixth bevel face **412** being connected to the third end face **413** at an end close to each other.

(38) The fifth bevel face **411** and the sixth bevel face **412** respectively extend from ends of two tooth faces of a second tooth **42**, and extending ends of the fifth bevel face **411** and the sixth bevel face **412** are close to each other and connected to the third end face **413**.

(39) FIG. 5 illustrates a relative position relationship between the gear **4** and the second guide tooth **3** and the first guide tooth **2** when the gear **4** initially engages the rack **1** from the first side **11** of the rack **1** in some embodiments.

(40) The principle of facilitating correct guiding of the gear **4** into the rack **1** by mutual cooperation of the first guide tooth **2** and the second guide tooth **3** is as follows: the third pointed shape portion **41** on a second teeth **42** of the gear **4** does not come into contact with the second guide tooth **3** and the first guide tooth **2** at the same time, and using front-back and upper-lower spatial relationships and geometric characteristics of the gear and the rack, the corresponding teeth interact when the angle of the corresponding teeth is appropriate, without being influenced by the other teeth, and when the corresponding teeth are just aligned and the gear **4** cannot be rotated, an adjacent tooth can intervene subsequently to break the just-aligned balance state, thereby allowing the guiding to continue.

(41) In some embodiments, the first guide tooth **2** is provided on a tooth end face of one of the first teeth **13**.

(42) In some embodiments, the top of the first guide tooth **2** and the top of the second guide tooth **3** being both flush with the tops of the first teeth **13**.

(43) In some embodiments, as shown in FIG. 3, the toothed portions engagement adaptation mechanism further includes a mounting member **5**, the mounting member **5** being movably provided on the first side **11**, and the second guide tooth **3** being provided on the mounting member **5**.

(44) In some embodiments, as shown in FIG. 1, the toothed portions engagement adaptation mechanism further includes an elastic element **6**, the elastic element **6** is provided between the mounting member **5** and the first side **11**.

(45) The mounting member **5** is kept at a second distance b from the rack **1** under an action of the elastic element **6**, that is, the second guide tooth **3** is kept at a second distance b from the rack **1** under the action of the elastic element **6**.

(46) In some embodiments, in the process where the gear **4** moves axially from the first side **11** toward the second side **12** of the rack **1** and is guided into the rack **1**, there are the following

situations.

(47) If the third end face **413** of one of the third pointed shape portions **41** of the gear **4** is located between the second end face **313** of the second pointed shape portion **31** and the first end face **213** of the first pointed shape portion **21**, then since the second guide tooth **3** comes into contact with the gear **4** earlier than the first guide tooth **2**, the gear **4** further passively rotates to a position of correct engagement with the rack **1** under the action of the second guide tooth **3**, thus achieving correct guiding of the gear **4** into the rack **1**.

(48) If the third end face **413** of one of the third pointed shape portions **41** of the gear **4** is aligned with the second end face **313** of the second pointed shape portion **31**, such that the gear **4** abuts against the second guide tooth **3** and cannot rotate, as an axial force on the gear **4** increases, the second guide tooth **3** under force compresses the elastic element **6** and moves toward the second side surface **12**, and when the second end face **313** of the second pointed shape portion **31** is further than the first end face **213** of the first pointed shape portion **21** from the gear **4**, the first end face **213** of the first pointed shape portion **21** starts to approach the fifth bevel face **411** or the sixth bevel face **412** of another third pointed shape portion **41** on the gear **4**, and turns the gear **4** by a certain angle after contact, such that the gear **4** rotates to a position of correct engagement with the rack **1**, thus achieving correct guiding of the gear **4** into the rack **1**.

(49) If the third end face **413** of one of the third pointed shape portions **41** of the gear **4** is aligned with the first end face **213** of the first pointed shape portion **21**, since the second guide tooth **3** comes into contact with the gear **4** earlier than the first guide tooth **2**, and the second end face **313** of the second pointed shape portion **31** is not aligned with the third end face **413** of another third pointed shape portion **41** on the gear **4** due to a geometric relationship, the second guide tooth **3** can still turn the gear **4** by a certain angle, such that the gear **4** rotates to a position of correct engagement with the rack **1**, thus achieving correct guiding of the gear **4** into the rack **1**.

(50) In some embodiments, as shown in FIG. 1, the toothed portions engagement adaptation mechanism includes a first shaft **7**, the first shaft **7** connecting the mounting member **5** and the rack **1**, the first shaft **7** extending parallel to a connecting line between the first side **11** and the second side **12**, the mounting member **5** being configured to translate relative to the rack **1**.

(51) Optionally, the mounting member **5** and the first shaft **7** move simultaneously relative to the rack **1**. Alternatively, the first shaft **7** does not move and the mounting member **5** moves along the first shaft **7** to move relative to the rack **1**.

(52) In some embodiments, the first side **11** is provided with a first recess **111**, the first recess **111** being recessed towards the second side **12**, the mounting member **5** being arranged in the first recess **111**, and the mounting member **5** including an outer side flush with the first side **11**.

(53) In some embodiments, the first guide tooth **2** is provided in the first recess **111**, and the mounting member **5** is provided with an avoidance portion **53** for avoiding the first guide tooth **2**.

(54) In some embodiments, the mounting member **5** is connected to the rack **1** by two first shafts **7**.

(55) As shown in FIG. 2, the first recess **111** is provided with two first holes **114**, the two first holes **114** being respectively used for the two first shafts **7** to pass through, and the two first shafts **7** connect the mount **5** and the rack **1**. The first recess **111** is also provided with a second hole **115**, the second hole **115** being used to mount the elastic element **6**.

(56) As shown in FIG. 3, the mounting member **5** is configured as a symmetrical structure, which includes two third holes **54** and two avoidance portions **53**. The two third holes **54** respectively correspond to the two first holes **114** on the rack **1**, and are used for the two first shafts **7** to pass through. The two third holes **54** are respectively formed at two ends of the mounting member **5** and are symmetrical with respect to the middle of the mounting member **5**. The second guide tooth **3** is provided between the two third holes **54**, with one avoidance portion **53** being formed between the second guide tooth **3** and one of the third holes **54**, and the other avoidance portion **53** being formed between the second guide tooth **3** and the other third hole **54**, and the two avoidance portions **53** being symmetrical with respect to the middle of the mounting member **5**. The mounting

member **5** configured as a symmetrical structure not only can be arranged on the rack **1** on one side of the logistics vehicle, but also can be arranged on the rack **1** on the other side of the logistics vehicle.

(57) In some embodiments, the mounting member **5** is fixedly connected to the two first shafts **7**, and the two first shafts **7** respectively pass through the two first holes **114** in the rack **1**, so the second guide tooth **3** and the two first shafts **7** can reciprocate together relative to the rack **1** in a direction of a connecting line of the first side **11** and the second side **12**.

(58) In some embodiments, the elastic element **6** is mounted between the first recess **111** and the mounting member **5** to provide a second distance **b** between the second guide tooth **3** and the first recess **111**.

(59) In some embodiments, as shown in FIG. **6**, the toothed portions engagement adaptation mechanism includes a second shaft **8**, the second shaft **8** connecting the mounting member **5** and the rack **1**, the second shaft **8** extending in a direction perpendicular to a connecting line between the first side **11** and the second side **12**, the mounting member **5** being configured to swing relative to the rack **1**.

(60) Optionally, the mounting member **5** is fixed to the second shaft **8**, and the mounting member **5** and the second shaft **8** both swing relative to the rack **1**. Alternatively, the second shaft **8** is fixed to the rack **1**, and the mounting member **5** swings around the second shaft **8** relative to the rack **1**.

(61) In some embodiments, as shown in FIG. **7**, the first side **11** is provided with a second recess **112**, and the mounting member **5** includes a first mounting portion **51** and a second mounting portion **52**, the first mounting portion **51** being arranged in the second recess **112** and the second guide tooth **3** being arranged in the first mounting portion **51**, the second shaft **8** connecting the second mounting portion **52** and the rack **1**. The elastic element **6** is arranged between the first mounting portion **51** and the second recess **112**.

(62) In some embodiments, the back face **14** of the rack **1** is provided with a third recess **113**, the third recess **113** being communicated with the second recess **112**, the second mounting portion **52** being arranged in the third recess **113**, the mounting member **5** including outer sides flush with the back face **14** and the first side **11** of the rack **1**.

(63) In some embodiments, as shown in FIGS. **6** and **7**, the second mounting portion **52** of the mounting member **5** is connected to the rack **1** by the second shaft **8**; the second guide tooth **3** is arranged on the first mounting portion **51** of the mounting member **5**; the elastic element **6** is arranged between the second recess **112** and the first mounting portion **51** to provide a second distance **b** between the second guide tooth **3** and the second recess **112**; and the rack **1** is machined with the first guide tooth **2**, which is adjacent to the position of the second guide tooth **3**, the first guide tooth and the second guide tooth being respectively located at the upper and lower sides of the axis of the gear **4** in height.

(64) The first guide tooth **2** and the second guide tooth **3** are respectively provided with the first pointed shape portion **21** and the second pointed shape portion **31** on the side of initial engagement with the gear **4**. When the third end face **413** of one of the third pointed shape portions **41** of the gear **4** is aligned with the second end face **313** of the second pointed shape portion **31**, the second guide tooth **3** swings around the second shaft **8** in a direction of approaching the second side **12**, and thus the first end face **213** of the first pointed shape portion **21** starts to approach the fifth bevel face **411** or the sixth bevel face **412** of another third pointed shape portion **41** on the gear **4**, and turns the gear **4** by a certain angle after contact, such that the gear **4** rotates to a position of correct engagement with the rack **1**, thus achieving correct guiding of the gear **4** into the rack **1**.

(65) According to the description of the above embodiments, the second guide tooth **3** may translate relative to the rack **1**, and may also swing relative to the rack **1**, so long as it can cause a change in the position of the second guide tooth **3** relative to the rack **1** such that the first guide tooth **2** engages the gear **4**.

(66) In some embodiments, the elastic element **6** includes a spring.

(67) In some embodiments, a logistics system is provided, which includes a logistics vehicle, a frame and the toothed portions engagement adaptation mechanism described above, wherein the gear **4** of the toothed portions engagement adaptation mechanism is provided on the logistics vehicle, and the rack **1** of the toothed portions engagement adaptation mechanism is provided on the frame, and the logistics vehicle is configured to move along the rack **1** by engagement of the gear **4** and the rack **1**.

(68) In some embodiments, the logistics system includes the toothed portions engagement adaptation mechanism in the above embodiments, and when the logistics vehicle switches from horizontal motion to vertical motion, the gear **4** can be smoothly guided axially into the rack **1** to achieve correct engagement.

(69) In some embodiments, the logistics system includes a third shaft **9** and a power mechanism provided on the logistics vehicle, wherein the third shaft **9** connects the power mechanism and the gear **4**, and the power mechanism is configured to drive the gear **4** to move from the first side **11** towards the second side **12** of the rack **1**.

(70) In some embodiments, each logistics vehicle is equipped with two or more sets of toothed portions engagement adaptation mechanisms, and the sets of toothed portions engagement adaptation mechanisms are symmetrically distributed on the left and right sides of the logistics vehicle. The rack **1** is fixedly mounted on a climbing rail (not shown in the figures) provided on the frame.

(71) In the description of the present disclosure, it needs to be appreciated that the use of the terms “first”, “second”, “third” and the like to define parts and components is only for the convenience of distinguishing the above-mentioned parts and components. Unless otherwise stated, the above terms have no special meanings, and therefore cannot be construed as limitations on the protection scope of the present disclosure.

(72) In addition, technical features of one embodiment may be beneficially combined with one or more other embodiments, unless explicitly denied.

(73) Finally, it should be noted that the above embodiments are only used for describing rather than limiting the technical solutions of the present disclosure. Although the present disclosure is described in detail with reference to the preferred embodiments, those of ordinary skill in the art should understand that they still can make modifications to the specific implementations in the present disclosure or make equivalent substitutions to part of technical features thereof; and such modifications and equivalent substitutions should be encompassed within the scope of the technical solutions sought for protection in the present disclosure so long as they do not depart from the spirit of the technical solutions of the present disclosure.

Claims

1. A toothed portions engagement adaptation mechanism, comprising: a rack comprising a first side and a second side arranged opposite to each other; a first guide tooth arranged on the first side; a second guide tooth movably arranged on the first side, the second guide tooth being far away from the second side relative to the first guide tooth; and a gear configured to partially engage the rack from the first side, and move towards the second side so as to completely engage the rack, wherein at least one of the first guide tooth and the second guide tooth is configured to engage the gear during the movement of the gear from the first side toward the second side.

2. The toothed portions engagement adaptation mechanism according to claim 1, wherein the second guide tooth is configured to move towards the second side under an action of an external force.

3. The toothed portions engagement adaptation mechanism according to claim 1, further comprising a mounting member, the mounting member being movably provided on the first side, and the second guide tooth being provided on the mounting member.

4. The toothed portions engagement adaptation mechanism according to claim 3, further comprising an elastic element provided between the mounting member and the first side.
5. The toothed portions engagement adaptation mechanism according to claim 3, further comprising a first shaft, the first shaft connecting the mounting member and the rack, the first shaft extending parallel to a connecting line between the first side and the second side, the mounting member being configured to translate relative to the rack.
6. The toothed portions engagement adaptation mechanism according to claim 3, wherein the first side is provided with a first recess, the first recess being recessed towards the second side, the mounting member being arranged in the first recess, and the mounting member comprising an outer side flush with the first side.
7. The toothed portions engagement adaptation mechanism according to claim 6, wherein the first guide tooth is provided in the first recess, and the mounting member is provided with an avoidance portion for avoiding the first guide tooth.
8. The toothed portions engagement adaptation mechanism according to claim 3, further comprising a second shaft, the second shaft connecting the mounting member and the rack, the second shaft extending in a direction perpendicular to a connecting line between the first side and the second side, the mounting member being configured to swing relative to the rack.
9. The toothed portions engagement adaptation mechanism according to claim 8, wherein the first side is provided with a second recess, and the mounting member comprises a first mounting portion and a second mounting portion, the first mounting portion being arranged in the second recess and the second guide tooth being arranged in the first mounting portion, the second shaft connecting the second mounting portion and the rack.
10. The toothed portions engagement adaptation mechanism according to claim 9, wherein a back face of the rack is provided with a third recess, the third recess being communicated with the second recess, the second mounting portion being arranged in the third recess, the mounting member comprising an outer side flush with the back face and the first side of the rack.
11. The toothed portions engagement adaptation mechanism according to claim 1, wherein the first guide tooth and the second guide tooth are respectively located on two sides of a central axis of the gear.
12. The toothed portions engagement adaptation mechanism according to claim 1, wherein there is a first distance between a part of the first guide tooth farthest away from the second side and a part of the second guide tooth farthest away from the second side, and the second guide tooth is configured to move towards the second side under an action of an external force, a maximum distance of the movement being a second distance, the second distance being greater than the first distance.
13. The toothed portions engagement adaptation mechanism according to claim 1, wherein a side of the first guide tooth away from the second side is configured as a first pointed shape portion, and a side of the second guide tooth away from the second side is configured as a second pointed shape portion, the first pointed shape portion and the second pointed shape portion both extending far away from the second side, and a part of the gear for initial engagement with the rack being configured as a third pointed shape portion, the first pointed shape portion and the second pointed shape portion being both adapted to the third pointed shape portion.
14. The toothed portions engagement adaptation mechanism according to claim 13, wherein the first pointed shape portion comprises a first bevel face, a second bevel face and a first end face, the first end face formed at a junction between the first bevel face and the second bevel face; and the second pointed shape portion comprises a third bevel face, a fourth bevel face and a second end face, the second end face formed at a junction between the third bevel face and the fourth bevel face.
15. The toothed portions engagement adaptation mechanism according to claim 1, wherein the rack is provided with two or more first teeth along a lengthwise extending direction, the first guide teeth

being provided on a side end face of one first teeth of the two or more first teeth.

16. The toothed portions engagement adaptation mechanism according to claim 1, wherein the rack is provided with two or more first teeth along a lengthwise extending direction, a top of the first guide tooth and a top of the second guide tooth being both flush with tops of the two or more first teeth.

17. A logistics system comprising a logistics vehicle, a frame and the toothed portions engagement adaptation mechanism of claim 1, wherein the gear of the toothed portions engagement adaptation mechanism is provided on the logistics vehicle, and the rack of the toothed portions engagement adaptation mechanism is provided on the frame, and the logistics vehicle is configured to move along the rack by engagement of the gear and the rack.

18. The logistics system according to claim 17, further comprising a third shaft and a power mechanism provided on the logistics vehicle, wherein the third shaft connects the power mechanism and the gear, and the power mechanism is configured to drive the gear to move from the first side towards the second side of the rack.
